University of Copenhagen / Department of Computer Science

Analyzing Greenland's Geochemistry through Machine Learning

Research Project Report

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Abstract

TO DO

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1 Introduction

Greenland is one of the world's most mineral-rich but least-explored regions. Over decades, the Geological Survey of Denmark and Greenland (GEUS) [1] has assembled an extensive public database of geochemistry samples, exploration reports, mineral occurrences, intrusions, geological maps, and license boundaries, all accessible through GEUS web services. This data is heterogeneous in coverage, quality, and completeness, which makes it an interesting and realistic setting for applying machine learning. The aim of this project is to use machine learning to analyse geochemical data from GEUS and explore what can be learned from it. For example, identifying patterns across elements, reconstructing missing measurements, mapping spatial trends, or relating geochemistry to known mineral occurrences.

1.1 Research questions

The project was organized around two major problems:

RQ1 — missing geochemistry. Given a sample predict an held-out element from the sample's other elements and its spatial neighborhood.

RQ2 — unexplored regions. Predict concentrations in an held out geological area rather than at random, avoiding spatial leakage.

This report focuses on **RQ1**, while setting some ground for the **RQ2**.

1.2 Contributions

This project applies one identical pipeline to two GEUS datasets with different sampling geometries. In this first part, I contribute with the following:

Contrasting different KNN integrations. Comparing distinct methods of encoding a spatial neighborhood as tabular data, across seven neighborhood sizes.

Evidence that spatial context is largely redundant. A strong baseline obtained only by a sample's own geochemistry doesn't gain much with geophysical or spatial information. Some elements are exceptions, however.

Comparison between hand-built vs learned neighbor aggregation. Two different graph neural networks (GNN), with distance-aware and unaware edges, are evaluated against the handbuilt KNN encodings.

2 Data and Methods

2.1 The datasets

Two public geochemical datasets, that differ sharply in sampling geometry (Figure 1 and table 1), were picked and pre-processed by the exact same code. Whole rock is an exploration sampling within a narrow band of South-West Greenland, whilst Stream Sediment covers the entire margin of Greenland.

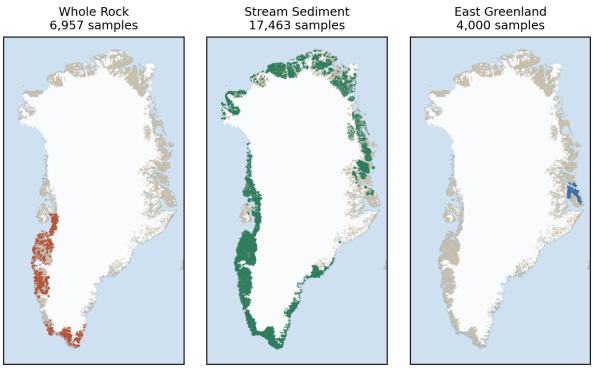


Figure 1: Sampling locations post-pre-processing on both datasets

Each source file is accompanied by a JSON meta file naming the sample number identifier, metadata, coordinates, and geophysical columns. The most important, geochemical columns, are whatever remains from this,

Table 1: The two datasets, before and after preprocessing.

	Whole rock	Stream sediment
Raw rows	26 068	20 158
Raw element columns	76	523
Merged rows	6 957	17 463
Merged element columns	76	93
Target elements (coverage $\geq$ 1 000)	48	71
(After pre-processing)		
Unique coordinate pairs	4 498	16 542
Co-located samples	3 458 (49.7 %)	1 759 (10.1 %)
Max. samples at one location	67	11
Longitude span	11.7°	59.5°
Latitude span	10.3°	23.8°

2.2 Preprocessing

The preprocessing of data consists on the following six steps:

1. **Mask below-detection values.** non-NaN entries that present negative values are usually interpreted as below-detection limit (BDL). Rather than keeping these values as is, or trying to perform educated approaches, BDL values are treated as NaN.

2. **Merge analytical columns.** Some columns represent a measurement of the same element but across different laboratories and methods. These different methods differ in detection limits, biases, instruments, etc, even if measured on the same sample for the same element, so they are aggregated by the median as a neutral design choice. This is what makes the Stream Sediment dataset move from 523 columns to 93.

3. **Merge by sample number.** Similarly to the step above, some rows are repeated across the same extracted sample, but measured by different methods. The aggregation is done in the exact same way: median across the rows depicting the same sample number.

4. **Log-transform.** After merging both dimensions, the numerical data is log_{10} transformed to follow the natural distribution of geochemistry. An added advantage is that this process diminishes the negative effect of outliers and sentinels.

5. **Coverage filter.** Element columns measured in fewer than 10 samples are completely dropped from the dataset, as columns with mostly NaN wouldn't add meaningful information for inference.

6. **Splits.** All experiments are done on the very same five-fold cross-validation. As we are still in the context of RQ1, there is no spatial holdout, and the samples are randomized, leading to a perfect 80/20 split in every fold (Figure 2).

2.3 Targets and evaluation

Every element measured in at least 1 000 samples is a target. Leading to 48 and 71 total targets on Whole Rock and Stream Sediment, respectively.

Each target is predicted from the sample's remaining elements and scored by R^2 over the five random folds. Samples that did not measure the target element are naturally discarded.

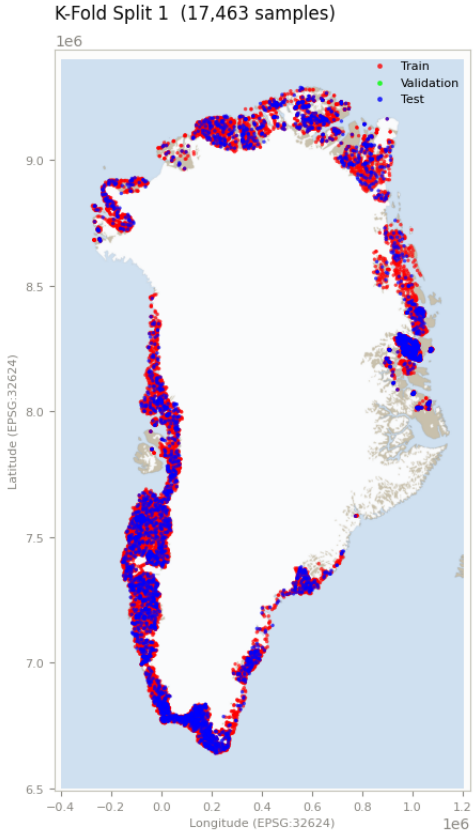


Figure 2: Example of sample distribution across splits in 5-Fold CV on the Stream Sediment dataset.

Once again, since we are in the context of RQ1, spatial leakage can be a problem due to lack of spatial CV. That is, a model trained on a certain sample can be guessing a sample just a few meters away, which can be considered duplicates in certain perspectives.

2.4 KNN Encoding

Neighbor features are built at training time, per fold. This means that a k-d tree is fitted on the training samples only, so a test sample never recognizes another test sample as a neighbor, which is a very straightforward method to block data leaks. Additionally, self-matches and co-located neighbors are discarded, meaning that a neighbor is necessarily at a different geographical location, not just a different sample from the same exploration site.

Four different types of encodings are compared. With k neighbours and m elements they differ in how much they aggregate (Table 2).

Table 2: The four neighbour encodings. Column counts are for $k = 20$, $m = 87$ (stream sediment).

Variant	Content	Added columns
A	Mean of the target element over k neighbours	1
B	The k individual target values	20
C	Mean of every element over k neighbours	87
D	Every element of *every* neighbour	1 740

2.5 Geophysical features

Geochemical neighbors only exist where somebody has already explored. Geophysical grids, on the other hand, are measured everywhere. Therefore, on unexplored ground, they are a motivated input.

Three geophysical attributes were added to each entry of both datasets: Magnetic anomaly, geothermal heat flow, and depth to Moho.

Despite wanting to introduce more geophysical features, I wasn't able to find more grids that properly covered all points in our dataset on the GEUS platform.

2.6 Models

Gradient-boosted decision trees remain one of the strongest machine learning methods on tabular data with this size and heterogeneity. The main model is the histogram-based gradient boosting (HGB) implementation in scikit-learn. As a comparison, a simple random forest (RF) and an XGBoost (XGB) model are also run on the same pipeline for the initial experiments, as to check that conclusions are not an artifact of model bias.

2.7 Graph neural network

A GNN offers a different way to use the sample neighborhood. Rather than aggregating by mean, or enumerating individual values into fixed columns, the relationships and their relevance is learned automatically. The graph presents one node per sample, and directed edges to the respective k nearest neighbors. So each sample sees the exact same neighborhood in the GNN as it saw in the hand-built version. Each edge carries $\log(1 + d)$, with d in kilometers. The node features are the sample's own elements. Since the network cannot consume NaN values, each element has two channels: A value channel where it is zero whenever the original value is NaN; And a presence/absence channel to serve as a flag.

Two different architectures are compared: A GraphSAGE, which is distance-blind by construction, and a graph attention network (GAT), that computes attention weights per edge from the distance, wich is something the previous KNN variants were missing. Other than this structural distinction, both use two message-passing layers of width 64, a skip connection from the node's features to a two layer head, dropout 0.1, and Adam optimizer at learning rate 10^{-2} with 10^{-4} weight decay.

GNN experiments were run only on Stream Sediment as it was the only dataset where standalone spatial signal was strong enough to justify a learnt aggregation.

3 Results

3.1 The geochemistry baseline is strong

Predicting a held-out element only from a sample's remaining elements reaches a mean R^2 of 0.7983 on whole rock (48 targets) and 0.8676 on stream sediment (71 targets). This is the baseline number every method from now on must improve upon. However, observing the results only by the mean R^2 can be misleading, as elements with more samples end up skewing it towards them. To cover that, in Table 3, the median and pooled variants are also presented. The first two metrics are pretty straightforward to understand and are computed over the several R^2 scores for each element. On the other hand, the latter one is the application of the R^2 formula over all samples for all elements.

Table 3: R^2 metrics in predicting an element through geochemistry alone, for both datasets, via RF, XGB, and HGB models

	Whole rock (48)			Stream sediment (71)		
R^2 Metric	RF	XGB	HGB	RF	XGB	HGB
Mean	0.7856	0.7954	**0.7983**	0.8587	0.8664	**0.8676**
Median	0.8076	0.8240	**0.8284**	0.8908	**0.9022**	0.9012
Pooled	0.7536	0.7642	**0.7678**	0.8135	**0.8259**	0.8244

3.2 Neighbor information should be pooled rather than enumerated

Table 4 gives the four encodings (Section 2.4) at $k = 20$ with HGB. In both datasets, the variant C has a much bigger relative gain when compared to the other methods, although it is not a significant difference by any means.

Table 4: The four KNN encodings on $k = 20$ and their respective mean R^2 and the respective difference to the baseline

	Whole rock (48)		Stream sediment (71)	
Variant	Mean R^2	Δ	Mean R^2	Δ
Baseline (no neighbours)	0.7983	—	0.8676	—
A	0.8045	+0.0062	0.8757	+0.0081
B	0.8046	+0.0063	0.8724	+0.0048
C	0.8125	**+0.0142**	0.8801	**+0.0125**
D	0.7980	−0.0003	0.8659	−0.0017

Sweeping k from 1 to 100 (Figure 3 and section A) reveals a pattern: On both datasets, variant A plateaus from $k \approx 5$ and variant C keeps improving until it flattens around $k = 50$; While variants B and D are consistently negatively affected as k rises. This constant decrease in performance is due to the fact that the dimensionality of the dataset in these variants is directly proportional to k. Variant D, the heavier one, has 1 740 columns at $k = 20$, while at $k = 100$, it already contains 8 700, with the possibility of many of these columns being NaN values. Essentially, even though variants B and D are more detailed about the neighborhood, the models are not capable of using it correctly and end up drowning in information.

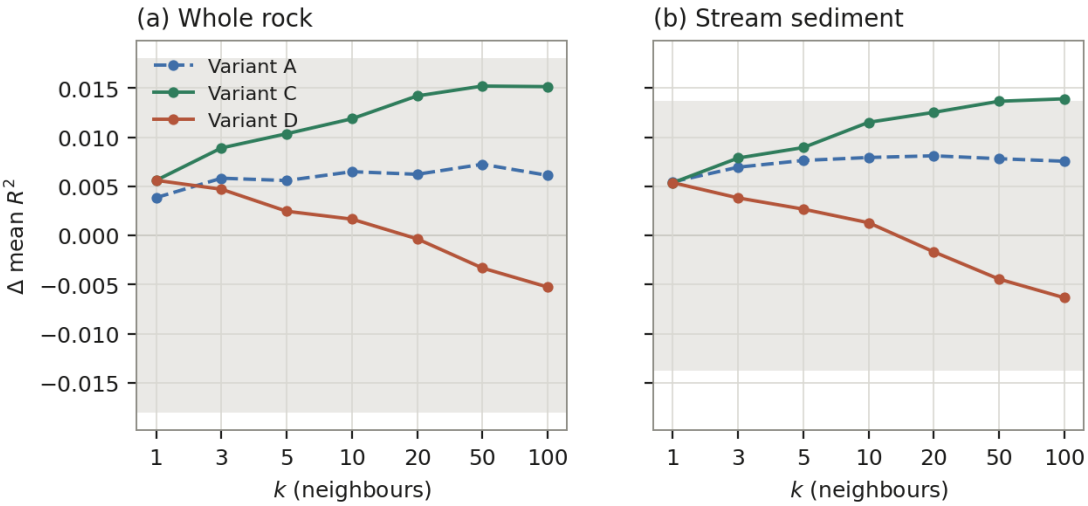


Figure 3: Mean R^2 as neighborhood k size varies on the full target set of each dataset. Transparent bands represent the fold standard deviation.

3.3 Relevance of each spatial information feature

By scoring each feature source on its own and in combination (Figure 4) allows us to understand what spatial information is more beneficial.

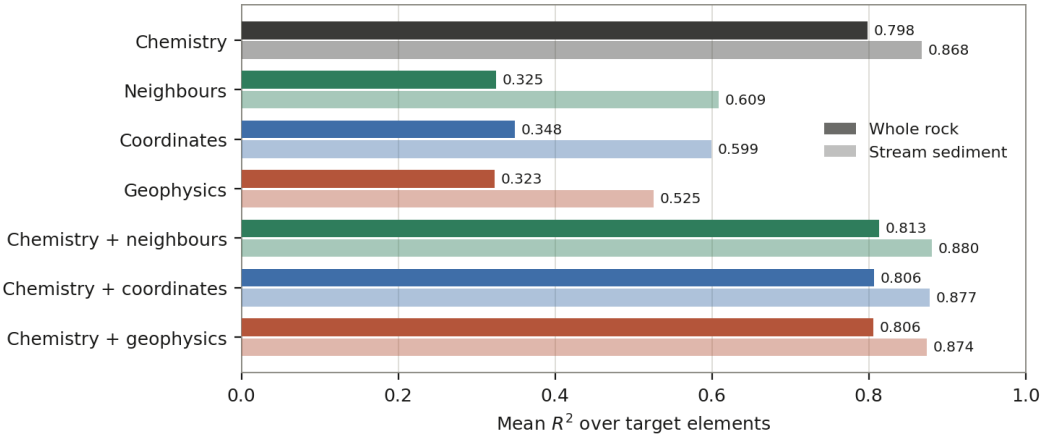


Figure 4: Mean R^2 by feature type over the respective target sets for both datasets.

Sampling geometry affects standalone spatial information Every standalone feature type is roughly twice as informative on stream sediment as on whole rock. This is a large difference that could be expected given the different sampling geometries of both datasets (Figure 1).

Sampling geometry does not affect the gain by spatial information. Even with such a different strength in standalone spatial signal, what that information adds to the current baseline is approximately the same: +0.015 versus +0.012 for neighbors, +0.008 versus +0.009 for coordinates, +0.008 versus +0.006 for geophysics. This means that the information ends up being redundant when faced with the strong geochemistry baseline, as a sample's own chemistry is a much stronger predictor than anything else.

Geophysics does not beat location. Coordinates mostly outscore geophysics on both datasets, and a combination of the two barely surpasses coordinates by itself (0.6017 versus 0.5994 on stream sediment). Essentially, the geophysics are serving as a proxy of location, encoding *where* a sample is rather than *what* is underneath. A model-independent correlation test also agrees to this result: by comparing the strongest correlation with a geophysical feature to the strongest in easting or northing, geophysics only wins on 27 of 64 (42%) elements of whole rock, and 26 of 81 (32%) on stream sediment. Some of the elements that more strongly resonate with geophysics rather than location can be seen in (Section B).

3.4 Elements with weaker baselines benefit from spatial information more

All R^2 metrics reported in the results above still make it hard to understand the improvement (or lack of) in different elements. By binning the targets based on their initial baseline score, we can properly verify that the elements with weaker initial scores do benefit more from additional data, especially regarding the surrounding neighborhood (Figure 5).

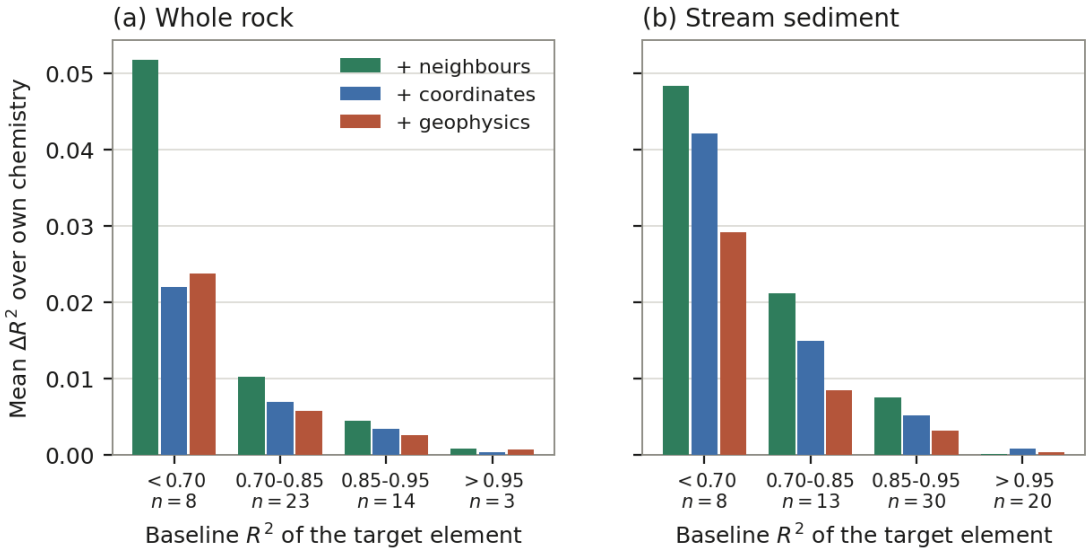


Figure 5: Mean improvement over own chemistry, with target elements binned by their baseline R^2 score (Section C)

Despite all the differences in these datasets, such as sample count, spatial extent, co-location rate, or targets, the decay is monotone, and the curves are nearly identical on both.

Neighbors are worth, on average +0.0517 to the eight whole-rock targets with a baseline < 0.70 and 0.0484 on stream sediment, with mean baseline of 0.5920 and 0.5846, respectively. However, with a baseline > 0.95, the increase in information only contributes 0.0008 for the three whole-rock targets and +0.0001 for stream sediment.

This result explains why any R^2 metric over such an extended target set can be misleading: Gain from neighbor information under 0.01 occurs in 40 out of 48 (83%) and 50 out of 71 (70%) target elements for both datasets, respectively, essentially omitting large gains for the few elements that end up benefiting.

3.5 Elements that benefit from spatial information are shared amongst datasets

The elements that benefit the most from the integration of neighbors and geophysics are not an arbitrary assortment, and are actually a similar recurrence in both datasets (Tables 5 and 6).

Table 5: The ten elements with the largest gain from neighbor features on each dataset, $k = 20$.

Whole rock		Stream sediment	
Element	$\Delta\ R^2$	Element	$\Delta\ R^2$
As	+0.096	U	**+0.221**
Au	+0.092	As	+0.060
Sb	+0.070	Br	+0.043
Cs	+0.048	Sb	+0.041
S	+0.041	Pb	+0.034
Pb	+0.027	Cs	+0.031
Cu	+0.026	Ca	+0.031
Rb	+0.023	Zn	+0.030
Sr	+0.018	Sr	+0.028
Ba	+0.017	Nb	+0.024

Table 6: The ten elements with the largest gain from geophysical features on each dataset

Whole rock		Stream sediment	
Element	$\Delta\ R^2$	Element	$\Delta\ R^2$
Au	+0.046	U	**+0.187**
As	+0.037	Br	+0.024
Cs	+0.027	Ca	+0.014
S	+0.027	Zn	+0.013
Sb	+0.019	As	+0.013
U	+0.017	S	+0.011
Cu	+0.015	Sr	+0.011
Fe2O3	+0.013	Ba	+0.011
Th	+0.012	Sb	+0.011
Ag	+0.011	Nb	+0.010

Arsenic, antimony, caesium, lead, and strontium are present in the top ten elements that benefit more from neighboring information on both datasets. The same can be said in the case of geophysical information for the elements arsenic, sulphur, antimony, and uranium. It is also worth noting that the latter element has a massive increase, both in relative and absolute terms, for the two feature sets on stream sediment data.

These results were obtained via different target sets, with different dimensions, on datasets with distinct geometry and characteristics. It should be, therefore, no coincidence that the same elements benefit from the same added features; however, the lack of proper geological knowledge prevents me from making any further comments or hypotheses regarding this.

3.6 Chemical isolation worsens the baseline

In Section 3.4, I reported on how elements with a weaker baseline tended to benefit more from additional features. However, what exactly leads to a lower baseline still needs to be discussed. As we are dealing with tabular algorithms, by nature, the closest correlations are used as leverage to find the correct approximated values. Likewise, a lack of such close correlation could mean the inability to perform. For each element, take the strongest (Spearman) correlation $|\rho|$ it has out of all the elements and call it *companion strength*. Essentially, how good of a proxy the sample already carries for a specific element.

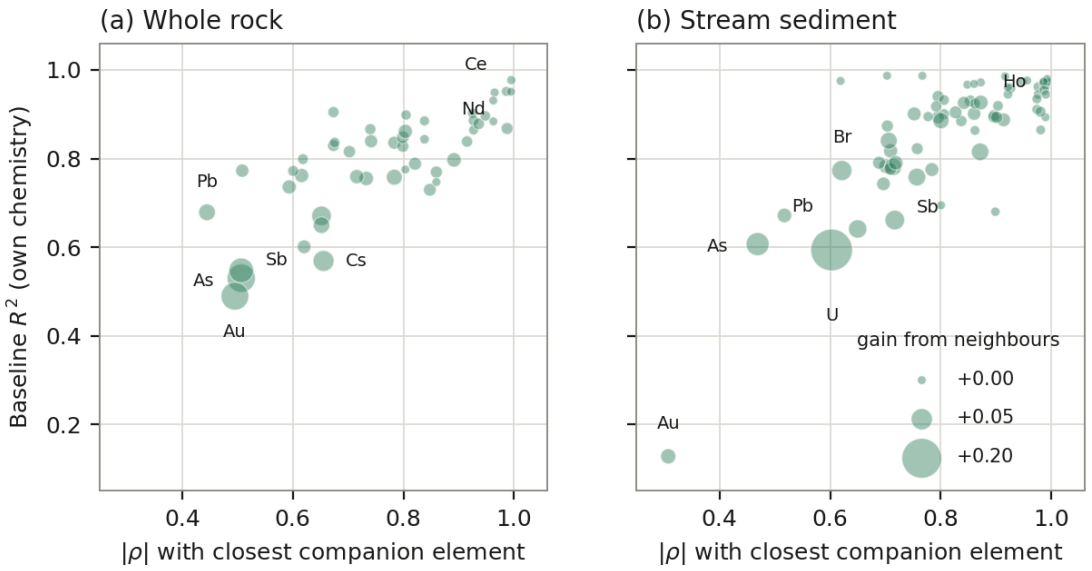


Figure 6: Baseline R^2 against companion strength. Area size defines the gain from neighbor integration.

This so-called companion strength fairly predicts the baseline on both datasets ($r = 0.77$ and 0.76, respectively)(Figure 6). Splitting the target elements by whether they have any partner above $\rho = 0.8$ into *isolated* and *accompanied* elements creates an interesting result shared among both datasets: Accompanied elements have a baseline of 0.8639 on whole-rock and 0.9200 on stream sediment, with a neighbor gain of 0.0047 on both, while isolated elements average 0.7380 and 0.7822 with a neighbor gain of 0.0242 and 0.0253, respectively. This increase in neighbor gain is between 5.2 to 5.4 times higher for accompanied than for isolated elements in both datasets, even though the split dimension is different (25 out of 48 (52%) and 27 out of 71 (38%) isolated), and the target elements differ.

What this exposes is a substitution. The elements at the top-right track each other almost perfectly (Ce–Pr $\rho = 0.995$, Ho–Er $\rho = 0.993$), and have several strong companions that are also accompanied elements. These elements are therefore largely predictable in the absence of the target element, making the neighborhood information unneeded. The elements at the lower-left have no such proxy: On stream sediment, for instance, gold's best companion is iron with $\rho = 0.307$, arsenic's is lithium at $\rho = 0.469$, sulfur's is mercury at $\rho = 0.517$, and none of these have a partner above $p \geq 0.80$ (Section E).

Tying together the previous sections, spatial context is worth something where the sample's own chemistry does not have a good proxy for the target.

3.7 A learned aggregation does not do better

Figure 7 compares both graph architectures against KNN-C across the same seven values of k, on identical folds. Note that contrary to Figure 3, this time the sweep is done over the full target set rather than the ten most measured elements.

The hand-built encoding wins everywhere. Variant C of the KNN integration is above both networks at all seven values of k, with no crossover at any point. The difference is not significant, however, especially when accounting for the standard deviation bands. Therefore, calling it a win is a big exaggeration. The plateau region that was reported in Section 3.2 is also similar to the region where the GNNs plateau.

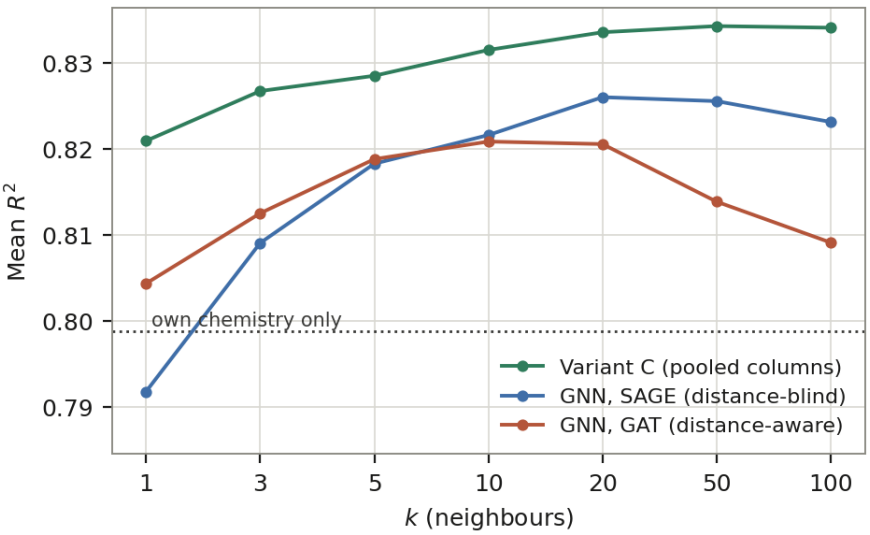


Figure 7: Mean R^2 on stream sediment for the two graph architectures and the hand-built KNN variant C encoding, across neighborhood size.

Distance-aware attention does not make a difference The two GNN architectures both follow similar results, with the best configuration being SAGE at $k = 20$ (0.8271). At a low k, the two methods discriminate the distance-weighted edges usefully, but as the neighborhood size increases, the difference in performance diminishes almost monotonically. After $k = 20$, the same effect that caused variant KNN-D to yield a lower performance seems to be taking action on the GNNs as well, albeit not to the same extent. This is because, just like variants B and D were drowning in information from tracking multiple columns, SAGE and GAT need to keep track of edges in the same manner.

4 Discussion

TO DO

5 Future work

The continuation into **RQ2** will be to hold out regions rather than samples. Essentially, instead of creating the train/test split randomly, it will be done through spatial block folds, isolating entire regions. For this, I plan to only work with Stream Sediment, as the Whole Rock dataset does not have as much coverage.

A K-Sweep Results

Table 7: Whole Rock: change over baseline.

k	A	B	C	D
1	0.0039	0.0039	0.0056	0.0056
3	0.0058	0.0053	0.0089	0.0047
5	0.0056	0.0063	0.0104	0.0025
10	0.0065	0.0061	0.0119	0.0017
20	0.0062	0.0063	0.0142	-0.0003
50	0.0072	0.0056	0.0152	-0.0033
100	0.0061	0.0051	0.0151	-0.0052

Table 8: Stream Sediment: change over baseline.

k	A	B	C	D
1	0.0055	0.0055	0.0054	0.0054
3	0.0069	0.0061	0.0079	0.0038
5	0.0076	0.0062	0.0089	0.0027
10	0.0079	0.0059	0.0115	0.0013
20	0.0081	0.0048	0.0125	-0.0017
50	0.0078	0.0024	0.0137	-0.0044
100	0.0076	0.0006	0.0139	-0.0063

B Individual geophysics gain over location

Element	**Best geophysical field**	**Correlation**	**Best location field**	**Correlation**	**Margin**
Se ppm	geo_heat_flow mW/m2	0.27976	(easting)	-0.069593	0.210168
Sn ppm	geo_depth_to_moho km	-0.602212	(northing)	-0.441925	0.160287
Au ppb	geo_heat_flow mW/m2	0.137533	(northing)	0.021406	0.116127
C wt%	geo_heat_flow mW/m2	-0.688319	(northing)	-0.582562	0.105757
Ga ppm	geo_depth_to_moho km	-0.334406	(northing)	-0.230134	0.104272
Br ppm	geo_depth_to_moho km	0.336817	(northing)	0.237108	0.099709
Sr ppm	geo_depth_to_moho km	-0.164298	(easting)	0.072871	0.091421
Ag ppm	geo_heat_flow mW/m2	0.123418	(easting)	0.032776	0.090642
La ppm	geo_magnetic_anomaly nT	0.115933	(northing)	-0.028659	0.087273
Ce ppm	geo_magnetic_anomaly nT	0.124151	(easting)	0.041162	0.082989

Table 9: Top 10 target elements with a wider margin between the best geophysical and location Spearman correlations on the Whole Rock dataset

Element	Best geophysical field	Correlation	Best location field	Correlation	Margin
Gd ppm	geo_heat_flow mW/m2	0.474911	(easting)	0.198662	0.276249
Al %	geo_depth_to_moho km	0.49066	(easting)	-0.263459	0.227201
SiO2 wt%	geo_depth_to_moho km	0.199027	(northing)	0.008254	0.190773
Na %	geo_depth_to_moho km	0.617319	(easting)	-0.50808	0.109239
Ca %	geo_depth_to_moho km	0.290458	(easting)	-0.193345	0.097112
Sm ppm	geo_depth_to_moho km	-0.339555	(northing)	-0.245418	0.094137
Dy ppm	geo_heat_flow mW/m2	0.5265	(easting)	0.439484	0.087016
Yb ppm	geo_depth_to_moho km	-0.446582	(easting)	0.368103	0.07848
S ppm	geo_depth_to_moho km	0.575251	(northing)	-0.499589	0.075662
Ce ppm	geo_depth_to_moho km	-0.22337	(northing)	-0.147771	0.075599

Table 10: Top 10 target elements with a wider margin between the best geophysical and location Spearman correlations on the Stream Sediment dataset.

C Information gain by baseline

Band	n	Baseline	+ Neighbours	+ Geophysics	+ Coordinates
under 0.70	8	0.5920	0.0517	0.0238	0.0219
0.70–0.85	23	0.7927	0.0102	0.0058	0.0070
0.85–0.95	14	0.8909	0.0044	0.0026	0.0034
over 0.95	3	0.9597	0.0008	0.0007	0.0003

Table 11: Model performance by baseline for whole rock samples.

Band	n	Baseline	+ Neighbours	+ Geophysics	+ Coordinates
under 0.70	8	0.5846	0.0484	0.0291	0.0421
0.70–0.85	13	0.7896	0.0211	0.0084	0.0150
0.85–0.95	30	0.9076	0.0075	0.0032	0.0051
over 0.95	20	0.9714	0.0001	0.0004	0.0009

Table 12: Model performance by baseline for stream sediment samples.

D Geophysical Correlation

Table 13: Whole Rock: top-10 elements by absolute Spearman correlation with magnetic anomaly (nT).

Element	n	Pearson	Spearman
C wt%	101	0.596	0.481
U ppm	3227	0.283	0.284
Zr ppm	3502	0.204	0.207
Nb ppm	1978	0.204	0.206
Co ppm	4378	−0.193	−0.197
Se ppm	305	−0.197	−0.193
Tm ppm	412	0.178	0.181
Ho ppm	433	0.132	0.177
FeO wt%	2490	−0.134	−0.169
Er ppm	444	0.120	0.166

Table 14: Whole Rock: top-10 elements by absolute Spearman correlation with heat flow (mW/m^2).

Element	n	Pearson	Spearman
C wt%	101	−0.780	−0.688
Sn ppm	443	0.497	0.503
Se ppm	305	0.264	0.280
Br ppm	715	−0.272	−0.268
Ge ppm	288	−0.184	−0.268
Er ppm	444	−0.252	−0.233
Tm ppm	412	−0.242	−0.222
W ppm	1815	0.166	0.208
Ho ppm	433	−0.233	−0.204
Dy ppm	447	−0.227	−0.177

Table 15: Whole Rock: top-10 elements by absolute Spearman correlation with depth to Moho (km).

Element	n	Pearson	Spearman
Sn ppm	443	−0.678	−0.602
C wt%	101	0.707	0.490
Br ppm	715	0.331	0.337
U ppm	3227	−0.268	−0.335
Ga ppm	2085	−0.323	−0.334
Cs ppm	1750	−0.263	−0.271
Nb ppm	1978	−0.223	−0.271
Rb ppm	4484	−0.218	−0.227
Lu ppm	3228	−0.219	−0.221
Se ppm	305	−0.184	−0.220

Table 16: Whole Rock: top-10 elements by absolute Spearman correlation with easting.

Element	n	Pearson	Spearman
C wt%	101	0.753	0.475
U ppm	3227	0.356	0.420
Co ppm	4378	−0.329	−0.312
Ni ppm	5526	−0.278	−0.298
W ppm	1815	−0.297	−0.291
As ppm	2355	0.182	0.267
Cs ppm	1750	0.261	0.256
Sb ppm	2102	0.171	0.245
Be ppm	972	0.222	0.244
MgO wt%	5545	−0.242	−0.241

Table 17: Whole Rock: top-10 elements by absolute Spearman correlation with northing.

Element	n	Pearson	Spearman
C wt%	101	−0.590	−0.583
Sn ppm	443	−0.437	−0.442
Nb ppm	1978	−0.441	−0.422
Ge ppm	288	−0.343	−0.381
U ppm	3227	−0.337	−0.380
Er ppm	444	−0.378	−0.336
Tl ppm	263	0.276	0.305
W ppm	1815	0.225	0.303
Tm ppm	412	−0.355	−0.296
Ho ppm	433	−0.332	−0.282

Table 18: Stream Sediment: top-10 elements by absolute Spearman correlation with magnetic anomaly (nT).

Element	n	Pearson	Spearman
S ppm	128	0.046	0.183
Fe ppm	130	−0.126	−0.177
Pr ppm	1402	0.156	0.171
Mn %	2118	−0.179	−0.170
Sn ppm	1739	0.154	0.158
Mg %	4041	0.121	0.147
Bi ppm	943	0.233	0.143
Sc ppm	13282	−0.080	−0.131
S %	3502	−0.091	−0.122
Cr ppm	14566	−0.093	−0.121

Table 19: Stream Sediment: top-10 elements by absolute Spearman correlation with heat flow (mW/m^2).

Element	n	Pearson	Spearman
Ho ppm	1394	0.391	0.532
Dy ppm	1401	0.379	0.526
Er ppm	1400	0.380	0.523
Tm ppm	1396	0.372	0.505
Gd ppm	1401	0.356	0.475
S ppm	128	0.290	0.469
Fe_2O_3t wt%	1330	0.406	0.418
Bi ppm	943	0.303	0.332
Au ppm	700	0.231	0.315
Ge ppm	1372	0.365	0.306

Table 20: Stream Sediment: top-10 elements by absolute Spearman correlation with depth to Moho (km).

Element	n	Pearson	Spearman
Be ppm	1722	0.681	0.699
Cd ppm	1750	0.659	0.663
Bi ppm	943	0.736	0.640
Ag ppm	2401	0.630	0.639
Ge ppm	1372	0.674	0.629
Na %	11692	0.495	0.617
S ppm	128	0.267	0.575
Al %	4041	0.494	0.491
Cl %	1125	0.418	0.488
W ppm	2492	0.361	0.448

Table 21: Stream Sediment: top-10 elements by absolute Spearman correlation with easting.

Element	n	Pearson	Spearman
Ge ppm	1372	0.820	0.759
Sn ppm	1739	0.646	0.719
Na %	11692	−0.378	−0.508
Tm ppm	1396	0.516	0.508
Se ppm	923	0.689	0.505
Er ppm	1400	0.479	0.486
Ho ppm	1394	0.474	0.469
S ppm	128	−0.282	−0.450
Dy ppm	1401	0.425	0.439
K %	10393	0.399	0.431

Table 22: Stream Sediment: top-10 elements by absolute Spearman correlation with northing.

Element	n	Pearson	Spearman
Ge ppm	1372	0.675	0.785
W ppm	2492	0.823	0.783
Bi ppm	943	0.807	0.781
Sn ppm	1739	0.558	0.772
Cd ppm	1750	0.708	0.752
Be ppm	1722	0.674	0.745
Se ppm	923	0.860	0.701
Ag ppm	2401	0.630	0.689
Na_2O wt%	10403	−0.750	−0.673
Au ppm	700	0.532	0.558

E Companion Strength

Table 23: Whole Rock: strongest companion elements by Spearman correlation.

Element	Coverage	Companion	Spearman	# partners > 0.8	Baseline
Ce ppm	3693	Pr ppm	0.995	5	0.977
Pr ppm	448	Ce ppm	0.995	10	–
Nd ppm	2936	Pr ppm	0.994	5	0.951
Y ppm	4109	Ho ppm	0.988	4	0.868
Ho ppm	433	Y ppm	0.988	7	–
La ppm	4188	Pr ppm	0.986	5	0.951
Er ppm	444	Ho ppm	0.985	7	–
Dy ppm	447	Ho ppm	0.968	7	–
Sm ppm	3764	Pr ppm	0.965	7	0.949
Tm ppm	412	Er ppm	0.964	6	–

Table 24: Whole Rock: weakest companion elements by Spearman correlation.

Element	Coverage	Companion	Spearman	# partners > 0.8	Baseline
Pb ppm	3429	Rb ppm	0.445	0	0.679
Bi ppm	719	Pr ppm	−0.470	0	–
Be ppm	972	Gd ppm	0.474	0	–
Au ppb	2389	Pr ppm	−0.495	0	0.489
Sb ppm	2102	As ppm	0.507	0	0.548
As ppm	2355	Sb ppm	0.507	0	0.530
Cd ppm	1008	Pr ppm	−0.509	0	0.773
Se ppm	305	Ag ppm	0.558	0	–

Table 25: Stream Sediment: strongest companion elements by Spearman correlation.

Element	Coverage	Companion	Spearman	# partners > 0.8	Baseline
Ho ppm	1394	Er ppm	0.993	8	0.973
Er ppm	1400	Ho ppm	0.993	8	0.979
Dy ppm	1401	Ho ppm	0.992	8	0.975
Nd ppm	12345	Pr ppm	0.990	5	0.965
Pr ppm	1402	Nd ppm	0.990	5	0.945
Y ppm	11857	Er ppm	0.990	5	0.893
Yb ppm	12469	Er ppm	0.987	5	0.955
Ce ppm	13213	Pr ppm	0.987	5	0.969
Tm ppm	1396	Yb ppm	0.986	8	0.973
La ppm	13300	Pr ppm	0.986	4	0.970

Table 26: Stream Sediment: weakest companion elements by Spearman correlation.

Element	Coverage	Companion	Spearman	# partners > 0.8	Baseline
Au ppb	2551	Fe ppm	0.307	0	0.128
Te ppm	227	Al %	0.348	0	–
Pd ppm	102	Ni ppm	0.399	0	–
As ppm	6873	Li ppm	0.469	0	0.607
S %	3502	Hg ppm	0.517	0	0.672
Hg ppm	758	Se ppm	0.520	0	–
U ppm	14831	Tm ppm	0.603	0	0.594
Au ppm	700	W ppm	0.613	0	–